

APPENDIX B - FAULT TREE ANALYSIS (FTA)

Aligned with Software General Architecture

Aligned FTA based on the upgraded FTA System document and the Software General Architecture document.

APPENDIX B

FAULT TREE ANALYSIS (FTA)

Drone-Remote Weapon Station

MIL-STD-882E System Safety Assessment Report Appendix

Document purpose	Provide qualitative FTA trees, hazard traceability, mitigations, and verification evidence in SAR-appendix form.
Source basis	MIL-STD-882E System Safety document, including PHA list, 40-hazard SHA log, verification procedures, and architecture notes.
FTA scope	One tree for each PHA hazard category
Method	Top-down qualitative fault tree analysis with OR / AND logic, transfer nodes, hazard-ID traceability, and mitigation / verification linkage.
Architecture emphasis	Explicit treatment of DRONE Flight Control <-> Turret Control <-> Recoil Mitigation couplings, plus power, sensor, comms, and retention paths.

Purpose and Scope

This appendix presents qualitative Fault Tree Analysis (FTA) artifacts for the drone-mounted remote weapon station in a format suitable for a MIL-STD-882E style Safety Assessment Report (SAR).

The appendix supplements the uploaded System Hazard Analysis by decomposing each major Preliminary Hazard Analysis (PHA) item into top events, intermediate events, and transferable basic-event groupings tied directly to the hazard log.

Source Material Used

The appendix was built from the source document sections covering the PHA list, the 40-hazard preliminary RAC table, the fail-safe CONOPS, the safety verification matrix, the SHA interaction diagram, the recoil-flight coupling analysis, the FMEA detection rationale, the cyber-safety interface, HFE / EMI requirements, and the fire-authorization chain.

FTA Conventions

Symbol / construct	Meaning	Use in this appendix
Top / intermediate event box	Hazardous state or logical contributor	Shown directly on each tree
AND gate	All conditions required	Used for combined failure conditions such as loss of link AND failed auto-DISARM
OR gate	Any condition sufficient	Used for alternative cause branches
Transfer node	Reference to detailed branch analysis table	Used to keep the figure readable while preserving traceability

FT-01 Unintended Fire

PHA hazard	Unintended fire	Related SHA hazard IDs	H-1, H-2, H-3, H-4, H-16, H-17, H-32, H-33, H-34, H-37
Top event	Weapon fires without a valid intentional and authorized fire sequence	Weapon fires without a valid intentional and authorized fire sequence	Weapon fires without a valid intentional and authorized fire sequence
Tree logic	OR	Primary appendix figure	Figure FT-01
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibit, and procedural barriers.	This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibit, and procedural barriers.	This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibit, and procedural barriers.

FT-01 Unintended Fire

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-1, H-32	SW logic fault; mode-handling error; inadvertent mode change	H-1: Fire-control SW defect, mode-handling error H-32: UI design; accidental click	H-1: Independent ARM state machine; dual-confirm (operator + guarded ARM); two-channel FIRE command ... H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-1: SW unit/integration tests; fault-injection; HIL bench test; H-32: Human factors test; UI validation
T2	H-2, H-16, H-37	Driver/relay stuck ON; trigger linkage or solenoid stuck; shot feedback hidden	H-2: Relay/MOSFET short, driver failure H-16: Mechanical trigger linkage jam; solenoid stuck H-37: Sensor disconnected	H-2: Series hardware inhibit (safety relay + mechanical sear/solenoid); monitored output feedback; H-16: Burst limiter; cut-off timer; redundant inhibit; monitored firing feedback (shot counter) H-37: Independent burst limiter; redundant detection (acoustic/accel optional); health monitoring	H-2: Bench electrical fault tests (short/open); output feedback validation; firing inhibit H-16: Burst-limit test; stuck-solenoid fault test; live-fire demo H-37: Sensor disconnect tests; burst limiter demo
T3	H-17, H-2, H-1	Failure to DISARM; safety relay/interlock stuck; BIT/watchdog ineffective	H-17: SW state latch; hardware interlock stuck H-2: Relay/MOSFET short, driver failure H-1: Fire-control SW defect, mode-handling error	H-17: Disarm in hardware + software; independent safety relay; disarm-on-fault; BIT; maintenance lock... H-2: Series hardware inhibit (safety relay + mechanical sear/solenoid); monitored output feedback; c... H-1: Independent ARM state machine; dual-confirm (operator + guarded ARM); two-channel FIRE command ...	H-17: Fault-injection; bench tests; safety relay verification H-2: Bench electrical fault tests (short/open); output feedback validation; firing inhibit t... H-1: SW unit/integration tests; fault-injection; HIL bench test; firing test with induced fa...
T4	H-3, H-33, H-34	Operator error; maintenance arming; transport actuation	H-3: Mode confusion; accidental trigger; UI ambiguity H-33: Procedure lapse; service mode missing H-34: Vibration; packed state	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-33: Maintenance lockout/tagout; physical safety pin; service mode disables firing circuits; SOP H-34: Transport lock; mechanical restraint; SAFE state; covers; shipping procedure	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-33: Maintenance procedure audit; lockout verification H-34: Transport vibration test; pack-out inspection

FTA assessment: This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibits, and procedural barriers. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-02 Loss of Communications While Armed

PHA hazard	Loss of comms armed	Related SHA hazard IDs	H-8, H-9, H-30, H-31, H-40
Top event	Weapon remains armed or controllable after command / video link loss or communications compromise	Weapon remains armed or controllable after command / video link loss or communications compromise	Weapon remains armed or controllable after command / video link loss or communications compromise
Tree logic	OR	Primary appendix figure	Figure FT-02
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state.	The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state.	The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state.

FT-02 Loss of Communications While Armed

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-8, H-40	RF link drop or jam occurs and auto-DISARM path fails	H-8: RF link drop/jam; integration fault H-40: Camera failure; link congestion	H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm;	H-8: Comm-loss tests (RF off/jam); timing verification; flight test H-40: Video loss tests; comm congestion tests;
T2	H-9	Weak authentication; spoofed packets; key compromise	H-9: Weak auth; key compromise; spoofed packets	H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-9: Pen-test; protocol tests; replay/spoof tests; code review
T3	H-40, H-8	Video lost while armed and inhibit FIRE without live video fails	H-40: Camera failure; link congestion H-8: RF link drop/jam; integration fault	H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l... H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; operator re-auth req...	H-40: Video loss tests; comm congestion tests; inhibit demo H-8: Comm-loss tests (RF off/jam); timing verification; flight test
T4	H-30, H-31	Software crash/reboot or time sync defect leaves unsafe state	H-30: Unhandled exception; memory leak H-31: Time drift; GPS time loss	H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller H-31: Monotonic timers; sanity checks; disable time-based gating on invalid time; alarm	H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo H-31: Time-drift tests; GPS-loss tests

FTA assessment: The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-03 Recoil-Induced Flight Instability

PHA hazard	Recoil instability	Related SHA hazard IDs	H-11, H-12, H-28, H-29, H-38
Top event	DRONE departs controlled flight during or immediately after weapon firing	DRONE departs controlled flight during or immediately after weapon firing	DRONE departs controlled flight during or immediately after weapon firing
Tree logic	OR	Primary appendix figure	Figure FT-03
Architecture focus	Matches SHA subsystem couplings and fail-safe barriers	Matches SHA subsystem couplings and fail-safe barriers	Matches SHA subsystem couplings and fail-safe barriers
Analysis note	This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks.	This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks.	This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks.

FT-03 Recoil-Induced Flight Instability

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-12	Damper leak/break; spring fracture; recoil cancellation degraded	H-12: Damper leak/break; spring fracture	H-12: Mechanical margins; end-stops; condition monitoring; scheduled inspection; inhibit sustained fi...	H-12: Endurance firing test; inspection; vibration test
T2	H-11, H-38	Turret control instability; encoder slip; unsafe burst / rate-of-fire	H-11: Stabilizer control fault; IMU failure H-38: Coupled dynamics; control mixing	H-11: Recoil mitigation; stabilization health monitoring; inhibit burst if stability out-of-bounds; r... H-38: Integration analysis; limit firing to stable hover; interlock with DRONE attitude bounds; tuning	H-11: Firing tests; IMU fault tests; stability threshold verification H-38: Flight tests; coupled-model simulation; firing in flight validation
T3	H-38	Flight-controller coupling; firing outside stable hover or attitude bounds	H-38: Coupled dynamics; control mixing	H-38: Integration analysis; limit firing to stable hover; interlock with DRONE attitude bounds; tuning	H-38: Flight tests; coupled-model simulation; firing in flight validation
T4	H-11	IMU failure/inconsistency; stabilization health monitor ineffective	H-11: Stabilizer control fault; IMU failure	H-11: Recoil mitigation; stabilization health monitoring; inhibit burst if stability out-of-bounds; r...	H-11: Firing tests; IMU fault tests; stability threshold verification
T5	H-28, H-29, H-19	Cable snag, EMI upset, or power competition perturb control loops	H-28: Routing error; wear H-29: RF emitters on DRONE; payload interference H-19: Battery drop; connector intermittent	H-28: Cable management; strain relief; clearance analysis; protective sleeving H-29: EMI shielding; filtering; separation; grounding; software robustness H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-28: Fit-check; vibration test; flight envelope inspection H-29: EMI tests; on-platform integration test H-19: Power interruption tests; connector wiggle tests

FTA assessment: This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-04 Turret Runaway Motion

PHA hazard	Turret runaway	Related SHA hazard IDs	H-6, H-7, H-18, H-27
Top event	Turret experiences uncontrolled yaw or pitch motion	Turret experiences uncontrolled yaw or pitch motion	Turret experiences uncontrolled yaw or pitch motion
Tree logic	OR	Primary appendix figure	Figure FT-04
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-homed logic.	This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-homed logic.	This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-homed logic.

FT-04 Turret Runaway Motion

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-6, H-7	Control loop instability; bad command; rate/position limiter failure	H-6: Control loop instability; sensor fault; encoder slip H-7: Control loop instability; IMU/encoder fault	H-6: Rate/position limiters; watchdog; torque/current limiting; mechanical stops; redundant encoder ... H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine	H-6: HIL stability tests; induced sensor fault; motion envelope tests H-7: Motion tests at extremes; fault injection; safe-park demo
T2	H-6, H-7, H-18	Encoder / IMU fault; slip; plausibility check ineffective	H-6: Control loop instability; sensor fault; encoder slip H-7: Control loop instability; IMU/encoder fault H-18: No homing; encoder offset	H-6: Rate/position limiters; watchdog; torque/current limiting; mechanical stops; redundant encoder ... H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings	H-6: HIL stability tests; induced sensor fault; motion envelope tests H-7: Motion tests at extremes; fault injection; safe-park demo H-18: Power-cycle tests; homing validation; safe-start demo
T3	H-27, H-7	Mechanical stop wear/fracture or safe-park routine fails	H-27: Stop fracture; wear H-7: Control loop instability; IMU/encoder fault	H-27: Stop strength margin; redundant limit switches; current limiting; periodic inspection H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine	H-27: End-stop overload test; inspection; motion tests H-7: Motion tests at extremes; fault injection; safe-park demo
T4	H-18	Startup homing/park sequencing defect causes unexpected motion	H-18: No homing; encoder offset	H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings	H-18: Power-cycle tests; homing validation; safe-start demo

FTA assessment: This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-home logic. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-05 Power Loss or Unsafe Re-Energization

PHA hazard	Power loss	Related SHA hazard IDs	H-19, H-20, H-30, H-36
Top event	Power anomaly causes loss of safety function or hazardous restart behavior	Power anomaly causes loss of safety function or hazardous restart behavior	Power anomaly causes loss of safety function or hazardous restart behavior
Tree logic	OR	Primary appendix figure	Figure FT-05
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing.	Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing.	Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing.

FT-05 Power Loss or Unsafe Re-Energization

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-19	Battery drop/connector intermittency and disarm-on-power-loss fails	H-19: Battery drop; connector intermittent	H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-19: Power interruption tests; connector wiggle tests
T2	H-20, H-36	Battery thermal runaway or actuator overcurrent causes electrical upset	H-20: Battery defect; overcurrent H-36: Stall; short; overheating motor	H-20: BMS protection; fusing; thermal isolation; venting; temperature monitoring; charging controls H-36: Current limiting; fuses; temperature monitoring; stall detection and shutdown	H-20: Abuse tests (as applicable); thermal monitoring validation; inspection H-36: Stall tests; thermal monitoring validation
T3	H-30, H-19	Crash/reboot recovery restores unsafe armed state	H-30: Unhandled exception; memory leak H-19: Battery drop; connector intermittent	H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo H-19: Power interruption tests; connector wiggle tests
T4	H-18, H-19, H-30	Restart bypasses re-authentication, SAFE initialization, or homing	H-18: No homing; encoder offset H-19: Battery drop; connector intermittent H-30: Unhandled exception; memory leak	H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller	H-18: Power-cycle tests; homing validation; safe-start demo H-19: Power interruption tests; connector wiggle tests H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo

FTA assessment: Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-06 Weapon / Ammunition Thermal Hazard

PHA hazard	Ammo cook-off	Related SHA hazard IDs	H-14, H-20, H-36
Top event	Overtemperature condition results in cook-off, fire, or thermal damage	Overtemperature condition results in cook-off, fire, or thermal damage	Overtemperature condition results in cook-off, fire, or thermal damage
Tree logic	OR	Primary appendix figure	Figure FT-06
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing.	This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing.	This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing.

FT-06 Weapon / Ammunition Thermal Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-14	High burst / blocked cooling occurs and thermal inhibit fails	H-14: High burst; blocked cooling	H-14: Thermal model; barrel temp sensor; firing duty-cycle limits; automatic cooldown inhibit; warnin...	H-14: Thermal tests; firing schedule validation; sensor calibration
T2	H-20	Battery defect / overcurrent bypasses BMS, venting, or thermal isolation	H-20: Battery defect; overcurrent	H-20: BMS protection; fusing; thermal isolation; venting; temperature monitoring; charging controls	H-20: Abuse tests (as applicable); thermal monitoring validation; inspection
T3	H-36	Motor stall/short and shutdown fails, causing secondary fire	H-36: Stall; short; overheating motor	H-36: Current limiting; fuses; temperature monitoring; stall detection and shutdown	H-36: Stall tests; thermal monitoring validation

FTA assessment: This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-07 Cyber / Jamming Hazard

PHA hazard	Cyber/jamming	Related SHA hazard IDs	H-8, H-9, H-29, H-31, H-40
Top event	Cyber or RF attack produces unauthorized control or defeats safety functions	Cyber or RF attack produces unauthorized control or defeats safety functions	Cyber or RF attack produces unauthorized control or defeats safety functions
Tree logic	OR	Primary appendix figure	Figure FT-07
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set.	This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set.	This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set.

FT-07 Cyber / Jamming Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-9	Spoofed / unauthorized command is accepted	H-9: Weak auth; key compromise; spoofed packets	H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-9: Pen-test; protocol tests; replay/spoof tests; code review
T2	H-8, H-40	Jamming or congestion removes control and SAFE transition fails	H-8: RF link drop/jam; integration fault H-40: Camera failure; link congestion	H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; operator re-auth req... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-8: Comm-loss tests (RF off/jam); timing verification; flight test H-40: Video loss tests; comm congestion tests; inhibit demo
T3	H-29, H-4	EMI / RF susceptibility corrupts control or inhibit lines	H-29: RF emitters on DRONE; payload interference H-4: EMI coupling into trigger line; ESD	H-29: EMI shielding; filtering; separation; grounding; software robustness H-4: EMI/ESD filtering; shielded cables; debounce + command authentication; redundant inhibit; groun...	H-29: EMI tests; on-platform integration test H-4: EMI/ESD tests (MIL-STD-461 as applicable); bench susceptibility tests
T4	H-31, H-9	Time validity / anti-replay checks defeated by drift or sync loss	H-31: Time drift; GPS time loss H-9: Weak auth; key compromise; spoofed packets	H-31: Monotonic timers; sanity checks; disable time-based gating on invalid time; alarm H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-31: Time-drift tests; GPS-loss tests H-9: Pen-test; protocol tests; replay/spoof tests; code review

FTA assessment: This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-08 Laser / Rangefinder Emission Hazard

PHA hazard	Laser hazard	Related SHA hazard IDs	H-21, H-22
Top event	Laser energy is emitted in an unsafe direction or at an unsafe time	Laser energy is emitted in an unsafe direction or at an unsafe time	Laser energy is emitted in an unsafe direction or at an unsafe time
Tree logic	OR	Primary appendix figure	Figure FT-08
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree.	Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree.	Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree.

FT-08 Laser / Rangefinder Emission Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-21	Operator uses laser in unsafe zone or without proper warning	H-21: Operator misuse; interlock failure	H-21: Laser class management; gated enable; interlock with aim zone; warning indicators; SOP; auto ti...	H-21: Laser safety test; interlock verification; operator training records
T2	H-21, H-22	Gated enable, interlock, or timeout fails	H-21: Operator misuse; interlock failure H-22: UI bug; stuck enable	H-21: Laser class management; gated enable; interlock with aim zone; warning indicators; SOP; auto ti... H-22: Positive enable + indicator; time-out; separate control path; BIT	H-21: Laser safety test; interlock verification; operator training records H-22: Functional tests; UI regression; emission enable tests
T3	H-22, H-32	UI / software bug leaves emission enabled when not intended	H-22: UI bug; stuck enable H-32: UI design; accidental click	H-22: Positive enable + indicator; time-out; separate control path; BIT H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-22: Functional tests; UI regression; emission enable tests H-32: Human factors test; UI validation

FTA assessment: Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-09 GUI / HMI Safety Failure

PHA hazard	GUI failure	Related SHA hazard IDs	H-3, H-10, H-32, H-39, H-40
Top event	Operator interface defect causes unsafe arming, firing, or control action	Operator interface defect causes unsafe arming, firing, or control action	Operator interface defect causes unsafe arming, firing, or control action
Tree logic	OR	Primary appendix figure	Figure FT-09
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture.	This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture.	This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture.

FT-09 GUI / HMI Safety Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-3, H-32	Ambiguous mode state; accidental mode change; poor annunciation	H-3: Mode confusion; accidental trigger; UI ambiguity H-32: UI design; accidental click	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-32: Human factors test; UI validation
T2	H-10, H-40	Latency, dropped packets, or video loss degrade operator picture	H-10: High latency; buffering; GUI lag H-40: Camera failure; link congestion	H-10: Time-stamped commands; aim-valid window; require stable track + operator confirm; inhibit if la... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-10: Network emulation tests; end-to-end latency tests; firing drills H-40: Video loss tests; comm congestion tests; inhibit demo
T3	H-39, H-23	Wrong datalink target / UI mix-up shown to operator	H-39: Wrong datalink target; UI mix-up H-23: EO/IR freeze; stale frames	H-39: HITL positive ID; ROE prompts; target labeling; confirm-before-fire; logs H-23: Stale-frame detection; sensor health; overlay warnings; inhibit auto-track on stale; operator c...	H-39: Operational scenario tests; training; UI validation H-23: Sensor fault injection; frame-freeze tests; field demo
T4	H-3, H-10, H-40	Confirm dialogs, alarms, or inhibit thresholds are bypassed	H-3: Mode confusion; accidental trigger; UI ambiguity H-10: High latency; buffering; GUI lag H-40: Camera failure; link congestion	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-10: Time-stamped commands; aim-valid window; require stable track + operator confirm; inhibit if la... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-10: Network emulation tests; end-to-end latency tests; firing drills H-40: Video loss tests; comm congestion tests; inhibit demo

FTA assessment: This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-10 Sensor Misidentification / Wrong-Target Engagement

PHA hazard	Sensor misID	Related SHA hazard IDs	H-23, H-24, H-25, H-39, H-40
Top event	System points or fires on the wrong target based on incorrect sensing or target identification	System points or fires on the wrong target based on incorrect sensing or target identification	System points or fires on the wrong target based on incorrect sensing or target identification
Tree logic	OR	Primary appendix figure	Figure FT-10
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence.	This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence.	This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence.

FT-10 Sensor Misidentification / Wrong-Target Engagement

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-23, H-40	EO/IR freeze, stale frame, or bad aim indication	H-23: EO/IR freeze; stale frames H-40: Camera failure; link congestion	H-23: Stale-frame detection; sensor health; overlay warnings; inhibit auto-track on stale; operator c... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-23: Sensor fault injection; frame-freeze tests; field demo H-40: Video loss tests; comm congestion tests; inhibit demo
T2	H-24	Auto-tracking drifts / mis-tracks under clutter	H-24: Algorithm error; clutter	H-24: HITL: require explicit operator confirm before FIRE; track confidence gating; easy drop-track c...	H-24: Tracking evaluation; scenario tests; operator-in-the-loop trials
T3	H-25	Ballistic solution wrong due to profile, zero, or wind input	H-25: Bad zero; wrong ammo profile; wind input	H-25: Zeroing procedure; ammo profile control; plausibility checks; display assumptions; inhibit if n...	H-25: Range zero tests; ballistic regression; configuration tests
T4	H-39, H-24	HITL positive ID / ROE / confirm-before-fire chain breaks down	H-39: Wrong datalink target; UI mix-up H-24: Algorithm error; clutter	H-39: HITL positive ID; ROE prompts; target labeling; confirm-before-fire; logs H-24: HITL: require explicit operator confirm before FIRE; track confidence gating; easy drop-track c...	H-39: Operational scenario tests; training; UI validation H-24: Tracking evaluation; scenario tests; operator-in-the-loop trials

FTA assessment: This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-11 Ammunition Feed Failure

PHA hazard	Feed failure	Related SHA hazard IDs	H-15, H-16, H-26, H-37
Top event	Feed subsystem failure creates unsafe firing behavior or inability to safely control fire	Feed subsystem failure creates unsafe firing behavior or inability to safely control fire	Feed subsystem failure creates unsafe firing behavior or inability to safely control fire
Tree logic	OR	Primary appendix figure	Figure FT-11
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards.	The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards.	The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards.

FT-11 Ammunition Feed Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-15	Feed jam occurs and trigger inhibit or jam-clear discipline fails	H-15: Feed chute damage; belt misalignment	H-15: Jam detection; immediate trigger inhibit; jam-clear procedure with DISARM required; robust feed...	H-15: Jam tests; maintainability demo; induced misfeed trials
T2	H-16	Trigger stuck or linkage jam creates continuing burst	H-16: Mechanical trigger linkage jam; solenoid stuck	H-16: Burst limiter; cut-off timer; redundant inhibit; monitored firing feedback (shot counter) to st...	H-16: Burst-limit test; stuck-solenoid fault test; live-fire demo
T3	H-37	Shot counter / feedback sensor fails and abnormal firing is hidden	H-37: Sensor disconnected	H-37: Independent burst limiter; redundant detection (acoustic/accel optional); health monitoring	H-37: Sensor disconnect tests; burst limiter demo
T4	H-26	Ammo count or belt state is wrong and operator decision is degraded	H-26: Sensor error; software counter drift	H-26: Dual method (shot count + belt sensor if available); reconcile on reload; warning on low confid...	H-26: Counter validation; reload scenario tests

FTA assessment: The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

FT-12 Structural / Retention Failure

PHA hazard	Structural failure	Related SHA hazard IDs	H-13, H-27, H-28, H-35
Top event	Structural or retention failure causes payload release, unsafe pointing, or loss of vehicle control	Structural or retention failure causes payload release, unsafe pointing, or loss of vehicle control	Structural or retention failure causes payload release, unsafe pointing, or loss of vehicle control
Tree logic	OR	Primary appendix figure	Figure FT-12
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers	Mapped to relevant command, actuation, sensing, power, and procedural barriers
Analysis note	This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes.	This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes.	This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes.

FT-12 Structural / Retention Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-13	Fastener loosening, fatigue, or overload breaks primary mount	H-13: Fastener loosening; fatigue; overload landing	H-13: Structural analysis with safety factors; locking fasteners; secondary retention; periodic torqu...	H-13: Static load + fatigue test; vibration test; inspection
T2	H-35	Latch / quick-disconnect improperly assembled or fails in service	H-35: Latch failure; improper assembly	H-35: Positive locking + secondary latch; assembly checklist; visual indicator; torque marks	H-35: Drop/load test; assembly verification; inspection
T3	H-27	Mechanical stop fracture or wear removes protective load limit	H-27: Stop fracture; wear	H-27: Stop strength margin; redundant limit switches; current limiting; periodic inspection	H-27: End-stop overload test; inspection; motion tests
T4	H-28	Cable routing / snag imposes side load or interference on structure	H-28: Routing error; wear	H-28: Cable management; strain relief; clearance analysis; protective sleeving	H-28: Fit-check; vibration test; flight envelope inspection

FTA assessment: This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

16. Hazard-Log Driven FTA Upgrade

Upgrade purpose. This section upgrades the FTA System appendix by explicitly aligning the fault-tree set to the current 40-hazard log. The upgrade adds coverage checks, RAC-driven prioritization, additional tree logic for hazards that were previously embedded or underrepresented, and closure evidence expectations for the Safety Assessment Report package.

16.1 Change Summary

- Added a full hazard-to-fault-tree coverage matrix for H-1 through H-40.
- Added RAC-driven FTA prioritization so RAC 1 and RAC 2 hazards receive explicit top-event or major branch treatment.
- Added four upgraded trees: FT-13 Authorized-Sector / Geofence Failure, FT-14 Software State-Machine Integrity Failure, FT-15 Maintenance / Transport Unsafe Actuation, and FT-16 Electrical / Actuator Thermal Escalation.
- Added minimal-cut-set style barrier checks for recurring safety barriers such as independent inhibit, heartbeat-to-SAFE, guarded operator action, sensor freshness, and power-fault SAFE behavior.
- Added FTA closure criteria so every tree has traceability to mitigation, verification evidence, residual-risk rationale, and open assumptions.

16.2 RAC-Driven FTA Treatment Rule

RAC group	FTA treatment	Expected evidence	Review emphasis
RAC 1 (High)	Dedicated top-event or explicit major branch. No closure by generic statement only.	Fault injection, HIL, operator evaluation, procedure audit, and safety-case signoff.	Authorization chain, human-in-the-loop confirmation, maintenance lockout, and loss-of-comms behavior.
RAC 2 (Serious)	Detailed transfer node with cause, barrier, and verification linkage.	Bench, HIL, environmental, cyber, or functional tests according to hazard type.	Independence of barriers and deterministic transition to SAFE.
RAC 3 (Medium)	Grouped tree branch acceptable if the branch preserves hazard ID traceability.	Inspection, motion envelope, endurance, regression, or integration evidence.	Failure detection and degraded-mode behavior.
RAC 4 (Low)	Traceability matrix entry plus monitoring/configuration control is acceptable.	Counter validation, time-drift tests, configuration checks, or operational log review.	Avoiding hidden common-cause contributors to higher-severity hazards.

17. Hazard Log to FTA Coverage Matrix

The table below is the controlling crosswalk between the hazard log and the FTA appendix.

Hazard ID	Hazard / unsafe condition	Initial RAC	FTA allocation	Primary FTA barrier / branch	Verification evidence to close FTA item
H-1	Unintended firing (SW logic fault)	RAC 2 (Serious)	FT-01 / FT-14	ARM state machine, dual-confirm, two-channel FIRE plausibility, watchdog/BIT.	SW unit/integration tests; fault-injection; HIL bench test; firing test with induced faults
H-2	Unintended firing (HW output stuck ON)	RAC 2 (Serious)	FT-01 / FT-15	Series hardware inhibit, output feedback, safe default on stuck output.	Bench electrical fault tests (short/open); output feedback validation; firing inhibit test
H-3	Unintended firing (operator error)	RAC 1 (High)	FT-01 / FT-09	Guarded controls, two-step FIRE, clear state annunciation, operator SOP.	Human factors review; UI simulation; operator eval; procedural compliance drills
H-4	Unintended firing (spurious EM/ESD)	RAC 2 (Serious)	FT-01 / FT-07	EMI/ESD hardening, debounce, authenticated command path, redundant inhibit.	EMI/ESD tests (MIL-STD-461 as applicable); bench susceptibility tests
H-5	Firing while turret not in authorized sector	RAC 2 (Serious)	FT-13	Sector/geofence inhibit, calibration check, hard stops, health monitoring.	Range test; calibration verification; limit-switch tests; firing inhibit demo
H-6	Turret runaway (yaw)	RAC 3 (Medium)	FT-04	Rate/position limiting, torque/current limiting, redundant encoder plausibility.	HIL stability tests; induced sensor fault; motion envelope tests
H-7	Turret runaway (pitch)	RAC 3 (Medium)	FT-04	Pitch hard stop, gravity compensation bounds, safe-park routine.	Motion tests at extremes; fault injection; safe-park demo
H-8	Loss of comms while armed	RAC 1 (High)	FT-02 / FT-07	Heartbeat supervision, auto-DISARM, local timer, re-authentication.	Comm-loss tests (RF off/jam); timing verification; flight test
H-9	Comm spoofing / unauthorized control	RAC 2 (Serious)	FT-02 / FT-07	Mutual authentication, signed commands, anti-replay, secure boot.	Pen-test; protocol tests; replay/spoof tests; code review
H-10	Latency / dropped packets causing wrong aim at FIRE moment	RAC 2 (Serious)	FT-02 / FT-09 / FT-13	Timestamped commands, aim-valid window, latency display, inhibit on stale data.	Network emulation tests; end-to-end latency tests; firing drills
H-11	Stabilization failure during firing	RAC 2 (Serious)	FT-03	Stability monitor, attitude bounds, burst inhibit, safe abort.	Firing tests; IMU fault tests; stability threshold verification
H-12	Recoil mitigation failure (mechanical)	RAC 3 (Medium)	FT-03	Mechanical margin, condition monitoring, inspection, sustained-fire inhibit.	Endurance firing test; inspection; vibration test
H-13	Mount structural failure	RAC 2 (Serious)	FT-12	Structural margin, locking fasteners, secondary retention, load tests.	Static load + fatigue test; vibration test; inspection
H-14	Weapon/munition overheating (cook-off)	RAC 2 (Serious)	FT-06	Thermal model, barrel-temperature sensing, duty-cycle / cooldown inhibit.	Thermal tests; firing schedule validation; sensor calibration
H-15	Ammunition box / feed chute jam	RAC 3 (Medium)	FT-11	Jam detection, immediate trigger inhibit, DISARM-required jam clearing.	Jam tests; maintainability demo; induced misfeed trials
H-16	Runaway burst (trigger stuck)	RAC 2 (Serious)	FT-01 / FT-11	Burst limiter, cut-off timer, firing feedback, redundant inhibit.	Burst-limit test; stuck-solenoid fault test; live-fire demo
H-17	Failure to DISARM (arm latch stuck)	RAC 3 (Medium)	FT-01 / FT-14	Hardware/software disarm, independent safety relay, BIT, lockout.	Fault-injection; bench tests; safety relay verification
H-18	Unexpected motion at power-up	RAC 2 (Serious)	FT-04 / FT-05	SAFE power-up, homing, park pose, inhibit until homed.	Power-cycle tests; homing validation; safe-start demo
H-19	Loss of power during armed state	RAC 2 (Serious)	FT-05 / FT-16	Fail-safe DISARM on power loss, state record, restart re-authentication.	Power interruption tests; connector wiggle tests
H-20	Battery thermal runaway / fire	RAC 2 (Serious)	FT-05 / FT-06 / FT-16	BMS, fusing, thermal isolation, venting, temperature monitoring.	Abuse tests (as applicable); thermal monitoring validation; inspection
H-21	Laser pointer hazard (eye exposure)	RAC 2 (Serious)	FT-08	Laser gated enable, aim-zone interlock, indicators, timeout.	Laser safety test; interlock verification; operator training records
H-22	Rangefinder emits when not intended	RAC 3 (Medium)	FT-08	Separate enable path, positive indicator, timeout, BIT.	Functional tests; UI regression; emission enable tests
H-23	Sensor failure leads to incorrect aim indication	RAC 2 (Serious)	FT-09 / FT-10 / FT-13	Stale-frame detection, sensor health, overlay warnings, auto-track inhibit.	Sensor fault injection; frame-freeze tests; field demo
H-24	Auto-tracking drifts / mis-tracks	RAC 2 (Serious)	FT-10 / FT-13	HITL confirmation, track confidence gate, drop-track control.	Tracking evaluation; scenario tests; operator-in-the-loop trials

H-25	Ballistic solution wrong	RAC 3 (Medium)	FT-10 / FT-13	Zeroing control, ammo profile control, plausibility checks, inhibit if not zeroed.	Range zero tests; ballistic regression; configuration tests
H-26	Ammunition count error	RAC 4 (Low)	FT-11	Shot count / belt-sensor reconciliation and confidence warning.	Counter validation; reload scenario tests
H-27	Mechanical stop failure	RAC 3 (Medium)	FT-04 / FT-12	Stop strength margin, redundant limits, current limiting, inspection.	End-stop overload test; inspection; motion tests
H-28	Cable snag/entanglement with DRONE	RAC 3 (Medium)	FT-03 / FT-12	Cable management, strain relief, clearance analysis, sleeving.	Fit-check; vibration test; flight envelope inspection
H-29	EMI causes control upset	RAC 3 (Medium)	FT-03 / FT-07 / FT-16	Shielding, filtering, separation, grounding, robust software.	EMI tests; on-platform integration test
H-30	Software crash / reboot during operation	RAC 2 (Serious)	FT-02 / FT-05 / FT-14	Safe reboot state, watchdog, state-machine recovery, redundancy.	Stress tests; fault-injection; long-run endurance; reboot recovery demo
H-31	Clock/time sync error affects command validity	RAC 4 (Low)	FT-02 / FT-07 / FT-14	Monotonic timers, sanity checks, alarm on invalid time basis.	Time-drift tests; GPS-loss tests
H-32	Inadvertent mode change (SAFE/ARM/AUTO)	RAC 2 (Serious)	FT-01 / FT-09 / FT-14	Guarded mode switch, confirmation, logging, audible/visual annunciation.	Human factors test; UI validation
H-33	Maintenance hazard – arming during service	RAC 1 (High)	FT-01 / FT-15	Maintenance lockout/tagout, physical safety pin, service mode inhibit.	Maintenance procedure audit; lockout verification
H-34	Transport hazard – accidental actuation	RAC 2 (Serious)	FT-01 / FT-15	Transport lock, mechanical restraint, SAFE state, covers, pack-out process.	Transport vibration test; pack-out inspection
H-35	Weapon release/quick disconnect failure	RAC 2 (Serious)	FT-12 / FT-15	Positive lock, secondary latch, checklist, visual indicator, torque marks.	Drop/load test; assembly verification; inspection
H-36	Overcurrent in actuator leads to fire	RAC 2 (Serious)	FT-05 / FT-06 / FT-16	Current limit, fuses, temperature monitoring, stall detection/shutdown.	Stall tests; thermal monitoring validation
H-37	Shot counter / feedback sensor failure hides firing	RAC 2 (Serious)	FT-01 / FT-11	Independent burst limiter, redundant firing detection, sensor health monitoring.	Sensor disconnect tests; burst limiter demo
H-38	DRONE flight controller interaction causes oscillations	RAC 2 (Serious)	FT-03	Coupled dynamics analysis, stable-hover interlock, attitude bounds, tuning.	Flight tests; coupled-model simulation; firing in flight validation
H-39	Friendly fire due to wrong target designation	RAC 1 (High)	FT-09 / FT-10 / FT-13	HITL positive ID, ROE prompts, target labeling, confirm-before-fire logs.	Operational scenario tests; training; UI validation
H-40	Loss of video feed while armed	RAC 2 (Serious)	FT-02 / FT-09 / FT-10 / FT-13	Auto-DISARM on video loss, fallback sensor/alarm, inhibit without live video.	Video loss tests; comm congestion tests; inhibit demo

18. FT-13 Authorized-Sector / Geofence Failure

PHA hazard	Firing outside authorized sector or after loss of valid aim context
Related SHA hazard IDs	H-5, H-10, H-23, H-25, H-39, H-40
Top event	System permits hazardous action when turret sector, aim validity, video freshness, or target designation is not valid
Tree logic	OR, with selected AND barrier-failure paths
Architecture focus	Hazard-log aligned FTA branch logic with mitigation and verification traceability
Analysis note	This tree closes the explicit H-5 gap and links geofence/sector authorization to sensor freshness, latency, ballistic validity, and HITL target confirmation.

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Primary mitigation / barrier	Verification / evidence
T1	H-5	Authorized sector is mis-set, stale, or calibration has drifted	Sector/geofence inhibit, hard stops, calibration check, health monitor	Range test, calibration verification, limit-switch tests, inhibit demo
T2	H-10, H-40	Aim command is accepted after excessive latency, dropped packets, or video loss	Timestamped command, aim-valid window, latency indicator, inhibit without live video	Network emulation, end-to-end latency, video-loss and congestion tests
T3	H-23	Sensor frame is stale or aim indication is incorrect	Stale-frame detection, sensor health, overlay warnings, auto-track inhibit	Sensor fault injection, frame-freeze test, field demo
T4	H-25	Ballistic or configuration input invalidates the authorized fire envelope	Zeroing procedure, controlled ammo profile, plausibility checks, inhibit if not zeroed	Range zero, ballistic regression, configuration tests
T5	H-39, H-24	Wrong target designation or automation mis-track passes operator confirmation chain	HITL positive ID, ROE prompts, target labeling, confirm-before-fire, track-confidence gate	Operational scenarios, tracking evaluation, operator-in-loop trials

FTA assessment: This tree closes the explicit H-5 gap and links geofence/sector authorization to sensor freshness, latency, ballistic validity, and HITL target confirmation. Closure requires evidence that each prevention barrier is independent where credited and that the failure response defaults to SAFE or DISARM for safety-relevant abnormal conditions.

19. FT-14 Software State-Machine Integrity Failure

PHA hazard	Software state, timing, or reboot defect defeats SAFE/ARMED/FIRING control assumptions
Related SHA hazard IDs	H-1, H-17, H-30, H-31, H-32
Top event	Software or state-machine defect permits unsafe armed state, unsafe command acceptance, or unsafe recovery
Tree logic	OR; selected contributors require command fault AND independent safety-monitor/inhibit failure
Architecture focus	Hazard-log aligned FTA branch logic with mitigation and verification traceability
Analysis note	This tree treats common software/state integrity as a cross-cutting FTA contributor instead of distributing it only inside individual event trees.

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Primary mitigation / barrier	Verification / evidence
T1	H-1	Fire-control software defect or mode-handling error creates invalid hazardous-action request	Independent ARM state machine, two-channel command plausibility, watchdog, BIT	Unit/integration tests, HIL, software fault injection
T2	H-17	Disarm latch or state variable remains armed after reset/fault	Hardware + software disarm, independent safety relay, BIT, maintenance lockout	Fault injection, bench tests, safety relay verification
T3	H-30	Crash/reboot recovers into unsafe or ambiguous state	Safe-state on reboot, watchdog, state-machine recovery, backup controller where justified	Stress, endurance, fault injection, reboot recovery demonstration
T4	H-31	Clock/time-sync error defeats freshness or anti-replay assumptions	Monotonic timers, sanity checks, alarm and disabled time-based gating when invalid	Time-drift and GPS-loss tests
T5	H-32	Inadvertent SAFE/ARM/AUTO transition due to UI or state synchronization error	Guarded mode switch, confirmation dialog, logging, annunciation	Human factors test and UI validation

FTA assessment: This tree treats common software/state integrity as a cross-cutting FTA contributor instead of distributing it only inside individual event trees. Closure requires evidence that each prevention barrier is independent where credited and that the failure response defaults to SAFE or DISARM for safety-relevant abnormal conditions.

19. FT-15 Maintenance / Transport Unsafe Actuation

PHA hazard	Unsafe arming, actuation, release, or discharge during service, handling, or transport
Related SHA hazard IDs	H-2, H-17, H-33, H-34, H-35
Top event	System actuates, remains armed, or releases payload during maintenance, transport, or assembly handling
Tree logic	OR, with AND logic where procedural lapse coincides with defeated physical lockout
Architecture focus	Hazard-log aligned FTA branch logic with mitigation and verification traceability
Analysis note	This tree elevates maintenance and transport hazards because H-33 is RAC 1 and H-34/H-35 are RAC 2 hazards requiring explicit evidence.

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Primary mitigation / barrier	Verification / evidence
T1	H-33	Service mode missing or lockout/tagout not enforced during maintenance	Maintenance lockout/tagout, physical safety pin, service-mode firing-circuit inhibit, SOP	Maintenance procedure audit and lockout verification
T2	H-34	Transport vibration or packed-state condition causes accidental actuation	Transport lock, mechanical restraint, SAFE state, covers, shipping procedure	Transport vibration test and pack-out inspection
T3	H-35	Quick-disconnect or weapon release latch is improperly assembled or fails	Positive locking, secondary latch, checklist, visual indicator, torque marks	Drop/load test, assembly verification, inspection
T4	H-2, H-17	Hardware output or arm/disarm interlock stuck in energized state during service	Series inhibit, monitored output feedback, independent safety relay, safe default	Bench electrical fault tests, output feedback, safety relay verification
T5	H-3, H-32	Operator or maintainer inadvertently changes mode while handling system	Guarded control, clear annunciation, confirmation, training and SOP	Human factors review and procedural compliance drills

FTA assessment: This tree elevates maintenance and transport hazards because H-33 is RAC 1 and H-34/H-35 are RAC 2 hazards requiring explicit evidence. Closure requires evidence that each prevention barrier is independent where credited and that the failure response defaults to SAFE or DISARM for safety-relevant abnormal conditions.

19. FT-16 Electrical / Actuator Thermal Escalation

PHA hazard	Electrical, power, EMI, or actuator fault escalates to loss of safety function, fire, or unsafe motion
Related SHA hazard IDs	H-19, H-20, H-29, H-36
Top event	Power or actuator electrical fault defeats safety monitoring or creates fire/thermal hazard
Tree logic	OR; common-cause branch when power fault also disables monitoring or inhibit path
Architecture focus	Hazard-log aligned FTA branch logic with mitigation and verification traceability
Analysis note	This tree complements FT-05 and FT-06 by consolidating electrical common-cause behavior across power, EMI, current, and thermal barriers.

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Primary mitigation / barrier	Verification / evidence
T1	H-19	Power interruption leaves unknown armed state or unsafe restart	Fail-safe DISARM on power loss, mechanical safe, state record, restart re-authentication	Power interruption and connector wiggle tests
T2	H-20	Battery defect or overcurrent causes fire or loss of drone	BMS protection, fusing, thermal isolation, venting, temperature monitoring	Abuse testing as applicable, thermal monitoring validation, inspection
T3	H-36	Actuator stall/short overheats motor or wiring	Current limiting, fuses, temperature monitoring, stall detection and shutdown	Stall tests and thermal monitoring validation
T4	H-29, H-4	EMI/ESD or RF payload coupling corrupts control, sensor, or inhibit line	Shielding, filtering, separation, grounding/bonding, robust parsing and debounce	EMI/ESD tests and on-platform integration tests
T5	H-30	Electrical transient induces controller crash/reboot and unsafe recovery	Watchdog, safe reboot, state-machine recovery, redundancy where justified	Stress, fault-injection, long-run endurance and reboot recovery demo

FTA assessment: This tree complements FT-05 and FT-06 by consolidating electrical common-cause behavior across power, EMI, current, and thermal barriers. Closure requires evidence that each prevention barrier is independent where credited and that the failure response defaults to SAFE or DISARM for safety-relevant abnormal conditions.

20. Cross-Cutting Minimal Barrier Checks

These are qualitative minimal-cut-set style checks. They do not assign probabilities; they identify combinations that must not be allowed to become single-point hazardous failures.

Cut-set family	Hazard IDs	Unsafe combination to prevent	Required independent barriers	Evidence needed
C1 Unauthorized fire command path	H-1, H-2, H-3, H-16, H-37	Invalid or unintended command reaches firing output AND independent inhibit does not block it.	Operator confirm, authenticated command, independent interlock, output feedback, burst limiter.	Interlock tests, stuck-output tests, HIL/fault injection, sensor-disconnect tests.
C2 Link/video loss while armed	H-8, H-40, H-10	Heartbeat/video lost or stale AND system remains armed or allows hazardous action.	Local timer, auto-DISARM, live-video/freshness inhibit, re-authentication.	RF-off/jam, video-loss, congestion, latency tests.
C3 Uncontrolled motion	H-6, H-7, H-18, H-27	Invalid command or sensor fault occurs AND torque/limit/stop barriers fail.	Rate/position limits, encoder plausibility, hard stops, torque disable, safe startup.	Motion envelope, limit-switch, homing, safe-park, induced sensor-fault tests.
C4 Wrong target / invalid aim context	H-5, H-23, H-24, H-25, H-39	Aim context invalid AND human confirmation or sector/track/ballistic gate fails.	HITL PID, ROE prompts, target labels, geofence, zero/profile control, stale-frame warning.	Scenario trials, tracking tests, configuration regression, range zero tests.
C5 Flight instability	H-11, H-12, H-28, H-29, H-38	Recoil or turret dynamics disturbance occurs AND stability/firing-envelope interlock fails.	Recoil mitigation, attitude bounds, stable-hover interlock, cable/EMI controls, safe abort.	Flight tests, coupled simulation, firing tests, vibration/EMI tests.
C6 Maintenance / transport actuation	H-33, H-34, H-35, H-17	Human/procedure lapse occurs AND physical/software lockout is ineffective.	Physical safety pin, service-mode inhibit, lockout/tagout, transport lock, secondary latch.	Procedure audit, lockout verification, transport vibration, assembly/drop/load inspection.
C7 Electrical thermal escalation	H-19, H-20, H-36, H-30	Power or actuator fault occurs AND shutdown/reboot/thermal protection fails.	BMS, fusing, current limit, thermal monitor, safe reboot, mechanical safe.	Power interruption, stall, thermal monitoring, reboot recovery tests.

21. FTA Verification Evidence Register

Evidence family	Hazard IDs supported	FTA trees supported	Acceptance intent	Example records
Fault injection / HIL	H-1, H-6, H-7, H-17, H-18, H-30	FT-01, FT-04, FT-05, FT-14	Injected software/sensor/timing faults force SAFE, inhibit, or controlled recovery.	HIL logs, fault reports, pass/fail test sheets.
Bench electrical / inhibit tests	H-2, H-16, H-17, H-36, H-37	FT-01, FT-11, FT-15, FT-16	Stuck outputs, open/short faults, and sensor disconnects do not produce uncontrolled hazardous output.	Electrical fault-test reports and output-feedback validation.
Communication / cyber tests	H-8, H-9, H-10, H-31, H-40	FT-02, FT-07, FT-09, FT-13	Loss, spoof, replay, latency, and malformed traffic result in rejection, alarm, or SAFE.	Protocol tests, replay/spoof reports, network-emulation logs.
Human factors / operator evaluation	H-3, H-32, H-39	FT-01, FT-09, FT-10, FT-13, FT-15	Operators can correctly identify mode, alarms, target identity, and required confirmation steps.	Usability test records, procedure drills, training records.
Environmental / EMI / vibration	H-4, H-12, H-13, H-28, H-29, H-34	FT-03, FT-07, FT-12, FT-15, FT-16	Physical and electrical disturbances do not defeat credited barriers.	MIL-STD-461 as applicable, vibration, fit-check and inspection reports.
Thermal / power tests	H-14, H-19, H-20, H-36	FT-05, FT-06, FT-16	Thermal and power faults lead to controlled inhibit, shutdown, or safe restart behavior.	Thermal logs, BMS validation, power interruption, stall tests.
Flight / coupled dynamics	H-11, H-38	FT-03	Vehicle remains within defined stable envelope or hazardous action is inhibited.	Coupled simulation, flight test, attitude-threshold verification.
Maintenance / assembly / transport	H-33, H-34, H-35	FT-15, FT-12	Handling conditions cannot arm, actuate, or release payload without defeating independent controls.	Lockout audit, pack-out inspection, drop/load test.

22. Open Items for Final Quantitative / Review Closure

Open item	Data required	Affected FTA areas	Reason for closure
Timing budget	Maximum allowed time for heartbeat loss, video loss, overcurrent, sensor fault, and software crash to transition to SAFE/DISARM.	FT-02, FT-04, FT-05, FT-13, FT-14, FT-16	Needed to judge deterministic safety response.
Independence evidence	Electrical and software separation evidence for ISM, hardware inhibit, safety relay, drive-enable, and mechanical safe barriers.	FT-01, FT-14, FT-15	Needed to avoid hidden common-cause barrier failure.
Sector and command envelope	Approved azimuth/elevation sector limits, rate/acceleration limits, calibration tolerance, and geofence definition.	FT-04, FT-13	Needed to close H-5 and motion runaway logic.
Flight-stability envelope	Permitted firing states, attitude/hover limits, recoil threshold, and coupling model validation.	FT-03	Needed to close recoil-flight instability branches.
Thermal limits	Barrel, actuator, battery, and electronics temperature thresholds plus cooldown logic.	FT-06, FT-16	Needed to validate thermal inhibit and shutdown branches.
Human factors evidence	Operator workload, mode annunciation, target-labeling, alarm priority, and maintenance procedure validation.	FT-01, FT-09, FT-10, FT-13, FT-15	Needed for HITL and maintenance hazards.

23. Software Architecture Alignment Upgrade

Upgrade purpose. This section aligns the upgraded FTA System document with the current Software General Architecture. The update converts software architecture controls into explicit FTA assumptions, software basic-event groups, tree branch updates, verification crosswalks, and closure criteria.

Alignment principle. The Independent Safety Monitor (ISM), fire / hazardous-action interlock, drive-enable gating, watchdogs, and fail-safe I/O are modeled as independent safety barriers. FCC, GUI, video, tracking, ballistic computation, logging, and telemetry may request, display, or record functions, but are not final authority for hazardous outputs.

23.1 Software Architecture Source Basis Used for FTA Alignment

Architecture topic	Software architecture position	FTA alignment impact
Safety authority separation	ISM operates independently from the FCC and enforces SAFE on communication loss, power fault, watchdog timeout, invalid command, or malfunction.	FTA trees involving hazardous motion, armed state, fire inhibit, or unsafe restart include an ISM/fail-safe branch.
Control versus safety partitioning	FCC and GUI generate functional requests; ISM validates requests before final drive-enable or hazardous-action output.	FTA shall not credit FCC/GUI alone as a safety barrier. Their failure is initiating or contributing unless stopped by ISM/interlock.
Software state machine	SAFE, STANDBY, ARMED, MOTION ENABLED, FAULT, and MAINTENANCE states with Any-State-to-SAFE transition.	FTA includes state-transition failures: failure to enter SAFE, unauthorized transition to ARMED, or recovery without authorized reset.
Interface integrity	OCU-FCC, FCC-ISM, drive, sensor, and maintenance interfaces use authentication, sequence, freshness, heartbeat, plausibility, and fail-safe rejection.	Communication, cyber, stale data, and replay branches map to interface-level controls.
Software verification matrix	V&V includes safe startup, comm-loss, invalid command, watchdog, limit enforcement, isolation, interlock independence, cyber stress, update controls, and recovery.	FTA closure identifies the software V&V evidence proving each credited software barrier.

23.2 Updated Architecture Assumptions for All FTA Trees

- The authoritative safety path is OCU/FCC request -> ISM validation -> interlock / drive-enable gate -> actuator or hazardous-output interface.
- The FCC is modeled as a functional command source, not as final safety authority.
- The GUI/HMI is modeled as display and operator-intent interface; it is not credited as an independent inhibit unless verified with ISM-authoritative status and interlock feedback.

- Loss, corruption, replay, delay, or ambiguity of any safety-relevant message is modeled as a basic event rejected by authentication, sequence/freshness checks, heartbeat supervision, or SAFE transition.
- Video, tracking, ballistic, and logging services are non-safety services whose failure must not block the ISM safety loop, torque disable, fire inhibit, or critical fault logging.
- Maintenance and update interfaces are separate contributors because unsafe configuration, unsigned firmware, or maintenance-mode lockout failure can create common-cause software hazards.

23.3 Software Architecture to FTA Allocation Matrix

Software element	Affected FTA trees	FTA basic-event family	Credited software control	Closure evidence
Operator GUI / HMI	FT-01, FT-08, FT-09, FT-10, FT-15	Mode confusion, stale status, wrong target display, accidental action, warning bypass.	Guarded controls, clear SAFE/ARMED/FAULT annunciation, stale data indication, alarm priority.	HMI review, operator-in-loop test, stale-data simulation, misuse-case test.
Communications Service	FT-02, FT-07, FT-09, FT-10, FT-14	Dropped packets, spoofing, replay, latency, malformed message, loss of heartbeat.	Authentication, sequence counter, freshness, heartbeat, strict parser, timeout to SAFE.	Network emulation, malformed-packet, replay/spoof, comm-loss timing.
FCC Control Services	FT-03, FT-04, FT-05, FT-10, FT-14	Invalid motion request, unstable control loop, bad recovery command, invalid mode request.	FCC cannot directly enable torque; requests are advisory until ISM validation.	HIL command-response logs, control-loop fault injection, interface review.
Independent Safety Monitor Firmware	FT-01 through FT-17	Failure to detect invalid command, failed watchdog, pass-through unsafe command, failure to enter SAFE.	Final safety authority, command validation, heartbeat supervision, torque disable, fail-safe default OFF.	ISM fault injection, watchdog, boundary command, torque-disable timing.
Fire / Hazardous-Action Interlock Logic	FT-01, FT-02, FT-06, FT-08, FT-11, FT-15	Hazardous output allowed without valid authorization or timeout.	Dual-channel authorization, independent inhibit, monitored output feedback, armed-state gating.	Interlock independence, dual-channel mismatch, output feedback validation.
Motor Drive / I/O Interface	FT-03, FT-04, FT-05, FT-12, FT-16	Uncontrolled motion, overcurrent, limit bypass, unsafe restart.	Separate enable path, hard torque-disable, limit enforcement, drive-fault monitoring.	Limit enforcement, stall/current, power-cycle, torque-off timing.
Video / Sensor Services	FT-02, FT-03, FT-08, FT-09, FT-10	Stale frame, frozen sensor, loss of live video, invalid stability data.	Stale detection, sensor plausibility, overlay warnings, inhibit use of invalid data.	Frame-freeze, sensor fault injection, latency measurement, stability validation.
Data Logger / BIT Service	FT-05, FT-14, FT-15, FT-17, FT-18	Startup BIT missed, evidence lost, storage full, recovery without traceable event.	Power-on/continuous/initiated BIT, critical event logging, fault code capture.	BIT validation, storage-full test, post-fault log review.
Maintenance / Update Interface	FT-15, FT-18, FT-19	Unauthorized update, unsafe configuration, maintenance lockout failure.	Signed packages, role access, audit trail, maintenance-mode lockout, rollback.	Unsigned package, rollback, access-control, maintenance lockout.

23.4 Software Basic-Event Library for FTA Reuse

Basic event ID	Basic event definition	Software owner	Requirement link	Typical tree usage
SW-BE-01	FCC issues invalid motion or mode request	FCC Control Services	SWR-SAFE-002 / SWR-SAFE-003	FT-03, FT-04, FT-14
SW-BE-02	ISM accepts command outside valid envelope	ISM Firmware	SWR-SAFE-003	FT-04, FT-13, FT-17
SW-BE-03	ISM fails to transition to SAFE on critical fault	ISM Firmware	SWR-SAFE-001	FT-02, FT-05, FT-14, FT-17
SW-BE-04	Interlock authorizes without valid dual channel	Interlock Logic	SWR-SAFE-004	FT-01, FT-08, FT-11
SW-BE-05	Drive enable remains active after fault	Motor Drive / I/O	SWR-SAFE-001 / 002	FT-04, FT-05, FT-16
SW-BE-06	Heartbeat timeout not detected or escalated	Comms / ISM	SWR-SAFE-001	FT-02, FT-07, FT-17
SW-BE-07	Malformed, replayed, or unauthorized command accepted	Comms Service	SWR-HZ-COMMS-01	FT-02, FT-07, FT-14
SW-BE-08	Latency or stale command treated as current	Comms / HMI	SWR-HZ-COMMS-01	FT-02, FT-09, FT-10
SW-BE-09	Stale/frozen video or sensor data not annunciated	Video / Sensor Services	SWR-HZ-SENSOR-01	FT-09, FT-10
SW-BE-10	Track confidence or ballistic/configuration plausibility not checked	Tracking / Ballistic Support	SWR-HZ-SENSOR-01	FT-10
SW-BE-11	Thermal/current/voltage threshold not detected	Health Monitoring	SWR-HZ-HEALTH-01	FT-05, FT-06, FT-16
SW-BE-12	Stability or attitude-bound inhibit not enforced	FCC / ISM / Sensor	SWR-HZ-HEALTH-01	FT-03
SW-BE-13	Power-up or reboot bypasses SAFE initialization	ISM / Boot / Recovery	SWR-HZ-RECOVERY-01	FT-05, FT-14

SW-BE-14	Maintenance mode permits hazardous output without lockout	Maintenance Interface	SWR-HZ-MAINT-01	FT-15
SW-BE-15	Software update/configuration authenticity not verified	Bootloader / Maintenance	SWR-SAFE-007	FT-19
SW-BE-16	Non-safety service starves safety loop	System Partitioning	SWR-SAFE-005	FT-17, FT-18
SW-BE-17	Fault log missing or insufficient	Logger / BIT	SWR-SAFE-006	FT-14, FT-18
SW-BE-18	Fault reset permitted without clearance/authorization	ISM / HMI	SWR-SAFE-008	FT-05, FT-14, FT-15

24. Alignment Updates to Existing FTA Trees

FTA tree	Software-alignment update	Software basic events to add	Software V&V evidence
FT-01 Unintended Fire	Add ISM + interlock as final authorization barrier. GUI/FCC command is an initiating request only.	SW-BE-04, SW-BE-07, SW-BE-13, SW-BE-18	Interlock independence; dual-channel mismatch; HMI misuse-case; watchdog-to-SAFE.
FT-02 Loss of Communications While Armed	Separate command-link heartbeat, video freshness, and FCC-ISM heartbeat. ISM local timer is credited safety authority.	SW-BE-03, SW-BE-06, SW-BE-08, SW-BE-09	Comm-loss timing, video-loss inhibit, re-authentication, log review.
FT-03 Recoil-Induced Flight Instability	Add stability-health branch using attitude bounds, sensor plausibility, and inhibit sustained fire if stability is out-of-bounds.	SW-BE-11, SW-BE-12, SW-BE-16	Stability threshold injection, coupled model/simulation, HIL/platform integration.
FT-04 Turret Runaway Motion	Add invalid FCC request rejected by ISM, drive-enable cannot be written directly by FCC, and hard torque-disable branch.	SW-BE-01, SW-BE-02, SW-BE-05	Command boundary, encoder/IMU fault injection, torque-disable timing.
FT-05 Power Loss or Unsafe Re-Energization	Add reboot recovery branch: default SAFE, state not restored to ARMED, authorized reset only after BIT/health checks.	SW-BE-03, SW-BE-13, SW-BE-18	Power-cycle, crash/reboot, recovery procedure, state persistence review.
FT-06 Weapon / Ammunition Thermal Hazard	Add health-monitor branch for duty-cycle, thermal/current thresholds, derate, inhibit, and cooldown annunciation.	SW-BE-11	Thermal threshold injection, cooldown inhibit validation, sensor calibration.
FT-07 Cyber / Jamming Hazard	Map to interface rules: authentication, sequence counters, freshness, parser rejection, cyber-stress isolation.	SW-BE-06, SW-BE-07, SW-BE-08, SW-BE-16	Replay/spoof, malformed-packet, cyber-stress safety-loop starvation.
FT-08 Laser / Rangefinder Emission Hazard	Treat emission enable as hazardous-action interlock path with timeout, positive enable indication, and maintenance lockout.	SW-BE-04, SW-BE-14	Emission enable test, UI regression, interlock verification.
FT-09 GUI / HMI Safety Failure	GUI is low safety authority and must display ISM-authoritative state; GUI is not final authority for fire or motion enable.	SW-BE-08, SW-BE-09, SW-BE-18	HMI validation, stale-state alarm, operator evaluation.
FT-10 Sensor Misidentification / Wrong-Target Engagement	Sensor/video services are non-safety; stale data, low confidence, and invalid config prevent hazardous state transition.	SW-BE-09, SW-BE-10	Frame-freeze, confidence gating, configuration/zeroing regression.
FT-11 Ammunition Feed Failure	Add feedback monitoring and independent burst limiter branch; logger records abnormal firing feedback.	SW-BE-04, SW-BE-11, SW-BE-17	Sensor disconnect, burst limiter, abnormal feedback log review.
FT-12 Structural / Retention Failure	Software support is monitoring, prompts, inspection logging, and limit enforcement; no software-only structural credit.	SW-BE-05, SW-BE-17	Inspection log, limit switch, maintenance checklist evidence.
FT-13 Authorized-Sector / Geofence Failure	Map sector limits to signed configuration, ISM command validation, calibration/BIT, and inhibit outside authorized sector.	SW-BE-02, SW-BE-15	Sector boundary, calibration verification, invalid configuration rejection.
FT-14 Software State-Machine Integrity Failure	Align to SAFE/STANDBY/ARMED/MOTION ENABLED/FAULT/MAINTENANCE and Any-State-to-SAFE transition.	SW-BE-03, SW-BE-13, SW-BE-17, SW-BE-18	State transition, fault injection, recovery authorization, log replay.
FT-15 Maintenance / Transport Unsafe Actuation	Add role authorization, service-mode lockout, signed diagnostics, audit trail, and physical lockout interface.	SW-BE-14, SW-BE-15, SW-BE-18	Maintenance-mode, lockout/tagout, audit log.
FT-16 Electrical / Actuator Thermal Escalation	Add current/temperature monitor, stall detection, drive fault input, and ISM torque-disable branch.	SW-BE-05, SW-BE-11	Stall/current, thermal validation, drive fault injection.

25. New FT-17: ISM Final Safety Authority Failure

Top event. Hazardous motion or hazardous output occurs because the final independent safety authority does not prevent, interrupt, or recover from an unsafe condition. Tree logic is OR at top level, with AND combinations where a functional request is unsafe AND the ISM/interlock barrier fails to reject or disable it.

Transfer node	Contributing event	Software cause basis	Required barrier	Verification evidence
T1	ISM internal fault or crash	ISM firmware task stops, memory/configuration error, internal self-test not effective.	External watchdog, fail-safe default OFF, BIT, minimal verified ISM code.	ISM crash injection, watchdog timeout, BIT fault insertion.
T2	Unsafe command pass-through	FCC produces command outside envelope and ISM accepts it.	Command range/rate/state validation, limit checks, counters, rejection and logging.	Boundary-value testing, invalid-command HIL injection.
T3	Failure to enter SAFE	Communication loss, watchdog timeout, power fault, limit fault, or sensor fault occurs but Any-State-to-SAFE fails.	Local ISM timers, hard torque-disable, FAULT state, latched reset policy.	Comm-loss, power interruption, limit fault, watchdog and sensor fault tests.
T4	Safety loop starvation/common cause	Video/logging/network load prevents timely safety-loop execution.	Task priority separation, resource limits, non-blocking logging, independent safety MCU where practical.	Resource exhaustion, cyber-stress, storage-full, video overload.
T5	Unauthorized recovery	Fault reset or reboot returns to ARMED/MOTION ENABLED without fault clearance and authorization.	Restart in SAFE, BIT before transition, authorized reset, re-authentication.	Crash/reboot recovery, authorized reset validation.

FT-17 assessment. This tree makes the common safety authority explicit and prevents hidden single-point reliance on the FCC, GUI, or communication link. Closure requires evidence that the ISM can force SAFE independently, including when non-safety services fail or become overloaded.

26. New FT-18: Software Partitioning / Common-Cause Resource Failure

Top event. A non-safety software service or shared computing resource prevents timely execution of a safety function, blocks safety-state transition, or hides safety evidence.

Transfer node	Common-cause branch	Failure mechanism	Software architecture control	Verification evidence
T1	Video pipeline overload	Video consumes CPU/bandwidth and delays safety messages or GUI updates.	Safety loop priority; resource partitioning; drop frames rather than delay safety.	Video overload and latency test.
T2	Logging/storage blocking	Logger blocks safety task or loses critical event sequence.	Non-blocking queue; reserve critical-event storage; storage-full warning.	Storage-full injection and log integrity review.
T3	Parser or network flood	Malformed traffic consumes resources or crashes parser.	Strict parser; rate limit; authentication before expensive processing; ISM independence.	Malformed/high-rate cyber-stress test.
T4	Shared clock/time fault	Time drift affects freshness, timeout, or anti-replay decisions.	Monotonic timers; sanity checks; reject time-dependent command when invalid.	Time-drift and GPS-loss tests.
T5	BIT diagnostic conflict	Maintenance diagnostic or initiated BIT interferes with operational safety functions.	Maintenance mode restriction; operational lockout; abort on authentication/integrity fault.	Maintenance diagnostics and mode transition review.

27. New FT-19: Software Update / Configuration Integrity Failure

Top event. Unauthorized, corrupted, stale, or incorrect software/configuration data changes safety behavior or invalidates FTA assumptions.

Transfer node	Configuration/update branch	Failure mechanism	Required control	Verification evidence
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T1	Unsigned or unauthorized firmware accepted	Bootloader activates unverified software.	Signed package, role control, release approval, rollback.	Unsigned/corrupted package tests; release audit.
T2	Incorrect safety configuration loaded	Sector limits, command envelope, calibration, or thresholds are wrong.	Signed configuration, version control, configuration BIT, change log.	Invalid config rejection; calibration/limit boundary test.
T3	Rollback not controlled	System reverts to older version without known fixes or compatible configuration.	Version compatibility, release tags, rollback approval.	Rollback test and compatibility review.
T4	Maintenance port misused	Unauthorized diagnostic or update session changes safety-relevant settings.	Role access, audit trail, restricted operational maintenance interface.	Access-control test and audit log review.
T5	Hazard/FTA evidence not updated after change	Software change modifies safety assumptions without hazard/FTA/V&V update.	Configuration control, safety-impact assessment, regression test.	Release note, traceability matrix, SAR review.

28. Software Requirement to FTA Traceability

Requirement ID	Requirement summary	FTA allocation	FTA logic usage	Closure evidence
SWR-SAFE-001	Enter SAFE on power-up, reset, watchdog timeout, communication loss, invalid command, or safety-monitor fault.	FT-02, FT-04, FT-05, FT-14, FT-17	Any-State-to-SAFE; default OFF; safe restart.	Fault injection, reset, comm-loss, watchdog, torque-disable tests.
SWR-SAFE-002	FCC shall not directly energize actuator torque or hazardous outputs without ISM validation and enable gating.	FT-01, FT-04, FT-11, FT-17	FCC failure is not sufficient if ISM/interlock works.	Architecture review, interface test, code review.
SWR-SAFE-003	ISM validates command range, rate, direction, current state, and limit status before motion enable.	FT-04, FT-13, FT-17	Invalid motion/sector command rejected.	Boundary-value and HIL tests.
SWR-SAFE-004	Hazardous-action commands require independent interlock authorization and clear operator state annunciation.	FT-01, FT-08, FT-09, FT-11, FT-15	Dual-channel interlock and HMI state branch.	Interlock and HMI review, misuse-case test.
SWR-SAFE-005	Safety-critical software separated from non-safety services.	FT-07, FT-09, FT-18	Non-safety overload cannot block SAFE.	Resource exhaustion and cyber-stress tests.
SWR-SAFE-006	Safety faults logged with time, state, source, and mitigation outcome.	FT-05, FT-11, FT-14, FT-18, FT-19	FTA evidence preserved for closure.	Post-fault log review.
SWR-SAFE-007	Software update and configuration loading verify authenticity and integrity before activation.	FT-13, FT-15, FT-19	Configuration/update cannot silently invalidate safety assumptions.	Signed-package, rollback, invalid config tests.
SWR-SAFE-008	Controlled recovery from FAULT requires fault clearance and authorized reset.	FT-05, FT-14, FT-15, FT-17	No automatic unsafe recovery.	Recovery procedure and reset authorization tests.
SWR-HZ-FIRE-01	Hazardous action requires valid state, operator confirmation, interlock, health, and no active inhibits.	FT-01, FT-08, FT-11, FT-15	Fire/emission/feed hazardous-output inhibit.	Dual-channel, HMI, inhibit and misuse-case tests.
SWR-HZ-MOTION-01	Reject motion outside position, rate, acceleration, limit, homing, power, encoder, or stability constraints.	FT-03, FT-04, FT-13, FT-16	Motion envelope and torque-off branch.	HIL envelope, limit switch, torque-off timing.
SWR-HZ-COMMS-01	Authenticate, sequence, freshness-check, and timeout-supervise safety messages.	FT-02, FT-07, FT-09, FT-10	Comms integrity branch.	Network emulation, replay/spoof, malformed-packet test.
SWR-HZ-HEALTH-01	Monitor thermal/current/voltage/stability/actuator/sensor health and derate/inhibit/SAFE as needed.	FT-03, FT-05, FT-06, FT-16	Health threshold branch.	Thermal/current/stability threshold validation.
SWR-HZ-SENSOR-01	Detect stale, invalid, uncalibrated, or low-confidence sensor/video/aim-assist data.	FT-09, FT-10	Sensor freshness and confidence branch.	Frame-freeze, calibration, scenario tests.
SWR-HZ-MAINT-01	Maintenance, transport, inspection, configuration, and diagnostic modes maintain lockout.	FT-12, FT-15, FT-19	Maintenance lockout and update/config branch.	Lockout/tagout, access-control and audit tests.
SWR-HZ-RECOVERY-01	After power loss, reboot, crash, or latched fault, restart in SAFE and require health checks and authorization.	FT-05, FT-14, FT-17	Recovery and re-authentication branch.	Power-cycle, crash/reboot and recovery testing.

29. Updated FTA Closure Criteria Aligned to Software Architecture

- Each FTA tree shall identify whether the credited safety barrier is ISM firmware, interlock logic, drive-enable hardware, HMI/operator action, communication integrity, health monitoring, BIT/logging, maintenance lockout, or physical/procedural control.
- Any branch that credits FCC, GUI, tracking, video, ballistic, or logging software shall state that the credited function is advisory, monitoring, indication, or evidence support only unless an independent ISM/interlock barrier is also present.
- Any branch that credits SAFE transition shall cite the trigger: power-up, reset, watchdog timeout, communication loss, invalid command, sensor fault, limit fault, power fault, ISM internal fault, or unauthorized maintenance/update condition.
- Any branch that credits command rejection shall specify the rejection basis: authentication, sequence/freshness, state compatibility, command envelope, limit status, sensor plausibility, health threshold, or configuration validity.
- Any branch that credits recovery shall require fault clearance, BIT/health pass, authorized reset, re-authentication where applicable, and safety log evidence.
- FTA closure remains open for any software-derived barrier until verification evidence is available and tied to the assessed software version/configuration baseline.

30. Software-Architecture Open Items for FTA Final Signoff

Open item	Data required	Affected FTA trees	Why it matters
Safety timing budget	Maximum allowed time from fault detection to torque-disable / SAFE output.	FT-02, FT-04, FT-05, FT-17	Without timing, watchdog and comm-loss branches cannot be closed.
Command envelope baseline	Approved pan/tilt position, speed, acceleration, sector, and rate limits.	FT-04, FT-13, FT-17	Required for ISM boundary testing.
Sensor redundancy policy	Single-channel versus redundant encoders, IMU, limit switches, shot feedback, temperature/current sensors.	FT-03, FT-04, FT-06, FT-10, FT-16	Required for common-cause and single-point exposure.
Fault reset and latching policy	Which faults latch, who may reset, and what evidence is required before reset.	FT-05, FT-14, FT-15, FT-17	Required for recovery branch closure.
Cybersecurity baseline	Authentication, encryption, signing, anti-replay, key management, and audit requirements.	FT-02, FT-07, FT-18, FT-19	Required to credit command integrity and update barriers.
Software version/configuration baseline	Assessed build ID, toolchain, configuration files, calibration/sector data, and release package.	All software-derived branches	Required so the FTA matches the tested system.
Independence evidence	Evidence showing ISM/interlock cannot be bypassed by FCC, GUI, video, tracking, or maintenance services.	FT-01, FT-04, FT-15, FT-17, FT-18	Required for safety authority separation claim.